Vopak

South Africa liquid fuels market story

Inland supply, corridor capture and tank turnover

Durban–Lesedi corridor into Gauteng

10 October 2026

Public evidence and illustrative capture cases. Customer access, operating limits and returns remain provisional.

Photo source: Royal Vopak | Terminal Durban | vopak.com



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HG's South Africa storyline

SITUATION | The market and its drivers	COMPLICATION | Trends begin to reverse	RESOLUTION | Vopak's outlook
Outline South African demand development and Vopak's market of interest: eastern South Africa within economic reach of its storage infrastructure and NMPP. Explain the demand drivers: greater diesel use in power generation; rail to road; fuel prices; more efficient vehicles; and slow GDP growth. Outline the supply picture: refinery closures, challenges at Secunda and closure of Mossgas. Show growth in imports and where they enter.	A number of trends that supported demand growth have started to reverse: road to rail; the end of loadshedding and spare generation capacity; EVs/HEVs; and higher fuel prices affecting kilometres travelled. Explore supply changes: a SAPREF restart; Secunda's coal-derived gas, MRG sales and potential extension to 2032; and Natref's future. Compare Durban storage–NMPP–Lesedi with road/rail from Durban, transport from Matola and transport from Walvis Bay.	Market share of Vopak in Durban and Lesedi Potential scope for new investment?

Source: HG's supplied storyboard, slides 30–32 (6 October 2026). Condensed framing; propositions and questions to investigate. The following page presents the current analytical response.

South Africa's fuel outlook and Vopak's investment case

SITUATION | An established import market	COMPLICATION | The outlook is changing	RESOLUTION | Capture before expansion
Demand remains substantial: 21.94bn litres of petrol and diesel sold in 2023. Petrol declined over 2013–2023 while diesel grew slowly. Refinery closures increased import dependence. In 2025, gross imports reached 16.70bn litres; Durban customs accounted for 13.16bn litres. Durban–NMPP–Lesedi connects imports to inland customers. Gauteng's 6.81bn-litre demand provides context (2022); delivered cost and access define Vopak's reachable market.	Demand drivers can reverse. Eskom's own OCGT fuel use fell 40% in FY2025. Rail recovery, EVs/HEVs and efficiency affect fuel demand; GDP and fuel prices affect activity and travel. Domestic supply can change. SAPREF, Secunda's gas/MRG choices and Natref require dated output scenarios; refinery capacity alone cannot determine imports. Vopak's route must compete with road/rail from Durban and supply via Matola or Walvis Bay on delivered cost, access and reliability.	Establish Vopak's current share and additional capture by product, customer and destination. Reconcile shared Durban–Lesedi transfers to count unique demand once. The new illustrative base gives Durban 2.63bn litres, including 1.32bn through Lesedi. It assumes 20% import capture and 50% onward routing; actual share remains open. Test working tanks, turns and dispatch first. New investment depends on secured incremental flows, a capacity shortfall and returns that withstand downside cases.

Progress: demand, port imports and operating evidence are assembled; capture cases and investment sensitivities are quantified. Next: validate customer flows, route economics and usable capacity.

Partner SCR: Henry storyboard, 6 Oct. Evidence: pp8–17 and 21–35; capture assumptions: p36; investment sensitivities: pp40–44. Years and scopes differ; capture is illustrative.

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Import dependence supports the terminal opportunity, while customer access determines Vopak capture

Baseline / reference	What the evidence establishes	What changes the outlook
Fuel demand and power	2023 Department sales: 21.94bn L. Petrol CAGR -2.08%; diesel +0.82%. Own Eskom fuel fell 40% in FY2025.	Sector activity, fleet replacement and power dispatch change fuel use. 2024 sales remain unverified.
Domestic supply and imports	2025 gross imports: 16.70bn L. Net imports: 15.15bn L. Durban customs: 13.16bn L (79%).	SAPREF and PetroSA restart paths, Secunda gas/MRG and Natref output can change domestic supply.
Corridor position and tank turnover	Shared MPP pipeline: 105 / 148ML per week in FY2025, 70.9% used. Vopak access remains unverified.	Route costs, access and customer capture determine flow; achievable turnover determines handling headroom.
Capture sensitivity and nine worlds	Base capture screen: Durban 2.63bn L; Lesedi 1.32bn L within that flow. Assumed shares, not measured volumes.	Capture cases are shown on p36. The nine-world investment illustration uses separate capture assumptions.

Department sales 2023; SARS customs 2025. Transnet Pipelines 2025 p8. Capture screen: 20% gateway share, 50% routed through Lesedi, two turns/month. Conditional illustration, no customer commitment.

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Existing assets provide the starting point for capturing demand and testing expansion

Working recommendation	Investment logic	Condition for commitment
Capture inland supply	Win accessible customer flows from imported and domestic sources.	Prove delivered competitiveness and sufficient customer volume.
Use existing assets first	The worked illustration favours existing assets in six worlds; access/dispatch improvement in two.	Confirm working capacity and the actual operating constraint.
Stage corridor and capacity spend	High demand / low domestic supply favours extra tanks in the illustration. Working-capacity assumptions materially affect its value.	Fund the least-cost viable option that removes the proven constraint.
Keep strategic options open	Test all nine worlds and emerging gateways; screen EV adjacencies separately.	Validate capture, chargeable revenues and capex. The illustrated tank choice reverses in the downside tests.

Working recommendation supported by an authored Lesedi illustration, not a calibrated forecast. Reported capacity is used; capture, working capacity, tariffs, capex and discount rate are assumptions. No customer volume is represented as secured.

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The revised working-capacity assumption changes the illustrated expansion value

Decision test	Numeric condition	ILLUSTRATIVE high / low case	Still needs verification
Customer volume	Tanks must outperform improving existing capacity.	Assumed capture 2.813bn L/year; tanks win above 2.765bn L/year.	No secured volumes inferred. Validate customer capture and contract duration.
Route economics	Delivered-cost headroom ≥ R0/m³.	R750 allowable less R650 assumed logistics = R100/m³ headroom.	Both prices are authored. Compare actual Durban, Maputo, Walvis Bay and domestic quotes.
Capacity constraint	Captured flow exceeds achievable handling.	Existing 1.860bn L/year; improvement 2.232bn; tanks 2.870bn.	124,000 m³ working estimate; 36,000 m³ estimated. Turns 1.25 / 1.50 remain assumptions.
Incremental return	Positive NPV and better value than the alternative.	Improvement NPV R+24m; combined R+82m. Difference R58m.	10% discount rate, 20 years, pre-tax. Actual tariffs, cost, tax and project schedule remain open.

ILLUSTRATIVE ONLY. Working tankage aligned to 124,000 m³. Difference was R8m at 126,000 m³; it is now R58m because the assumed rental allocation changes. Commercial terms and engineering remain unvalidated. Details pp40–43.

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Only one illustrative world favours tanks; six favour existing assets and two favour improvement

Gauteng demand / domestic supply	HIGH deliveries 4.8bn L/year	MEDIUM deliveries 3.6bn L/year	LOW deliveries 2.4bn L/year
LOW demand 6.0bn L/year	**L / H** **USE EXISTING ASSETS** **Capture 0.99bn L/year** **Incremental NPV R+0m**	L / M USE EXISTING ASSETS Capture 1.50bn L/year Incremental NPV R+0m	L / L USE EXISTING ASSETS Capture 2.01bn L/year Incremental NPV R+0m
MEDIUM demand 6.8bn L/year	M / H USE EXISTING ASSETS Capture 1.39bn L/year Incremental NPV R+0m	**M / M | REFERENCE** **USE EXISTING ASSETS** **Capture 1.90bn L/year** **Incremental NPV R+0m**	M / L IMPROVE ACCESS / DISPATCH Capture 2.41bn L/year Incremental NPV R+24m
HIGH demand 7.6bn L/year	H / H USE EXISTING ASSETS Capture 1.80bn L/year Incremental NPV R+0m	H / M IMPROVE ACCESS / DISPATCH Capture 2.30bn L/year Incremental NPV R+24m	**H / L** **TEST IMPROVEMENT + TANKS** **Capture 2.81bn L/year** **Incremental NPV R+82m**

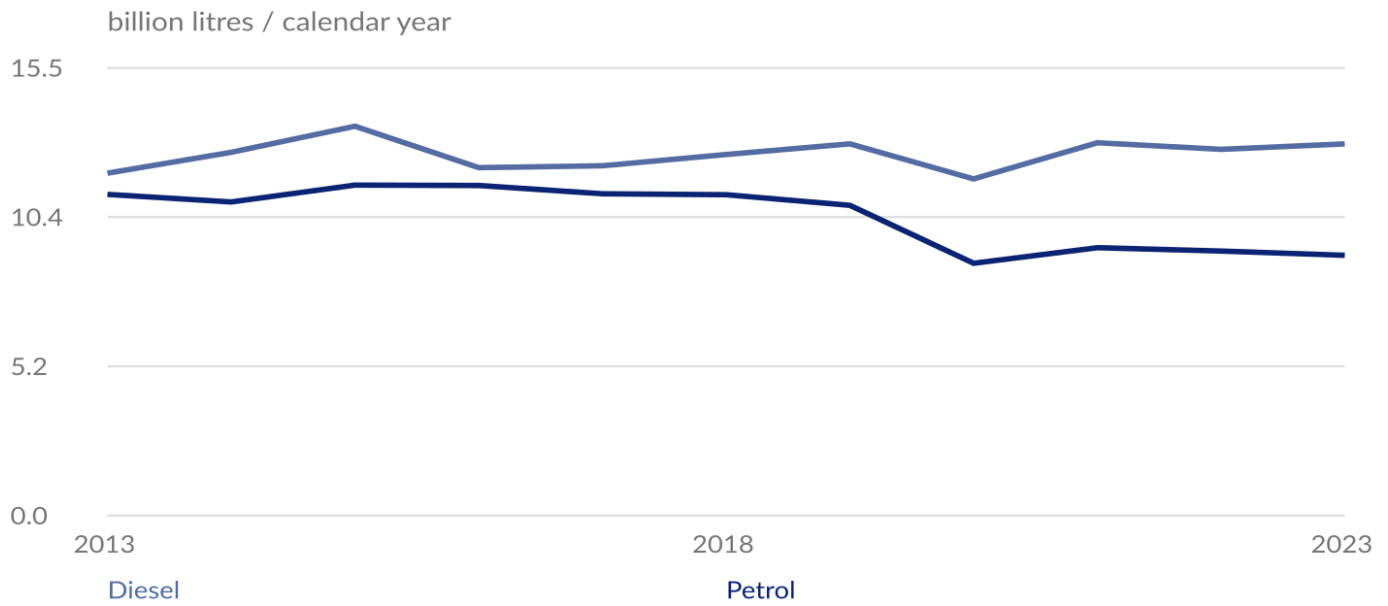
Highest incremental NPV among three illustrative packages; R0 means retain existing assets, not zero terminal value.

ILLUSTRATIVE snapshot, not forecast or probabilities. Domestic supply here means assumed deliveries into Gauteng, not national refinery capacity. Shares, rates and costs held fixed across worlds. Rail/EV opportunities remain separate cases, not valued here. Each cell is relative to existing assets; unserved volumes can remain. Inputs: p42; downside: p43.

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Petrol demand declined while diesel grew slowly over 2013–2023

Demand trend



FIASA 2024 observations are excluded from this comparison: source definitions are not reconciled.

Change over 2013–2023

01 | Petrol 2013–2023

11.15 to 9.04 bn litres
-2.12 bn L / -19.0%
CAGR -2.1% per year

02 | Diesel 2013–2023

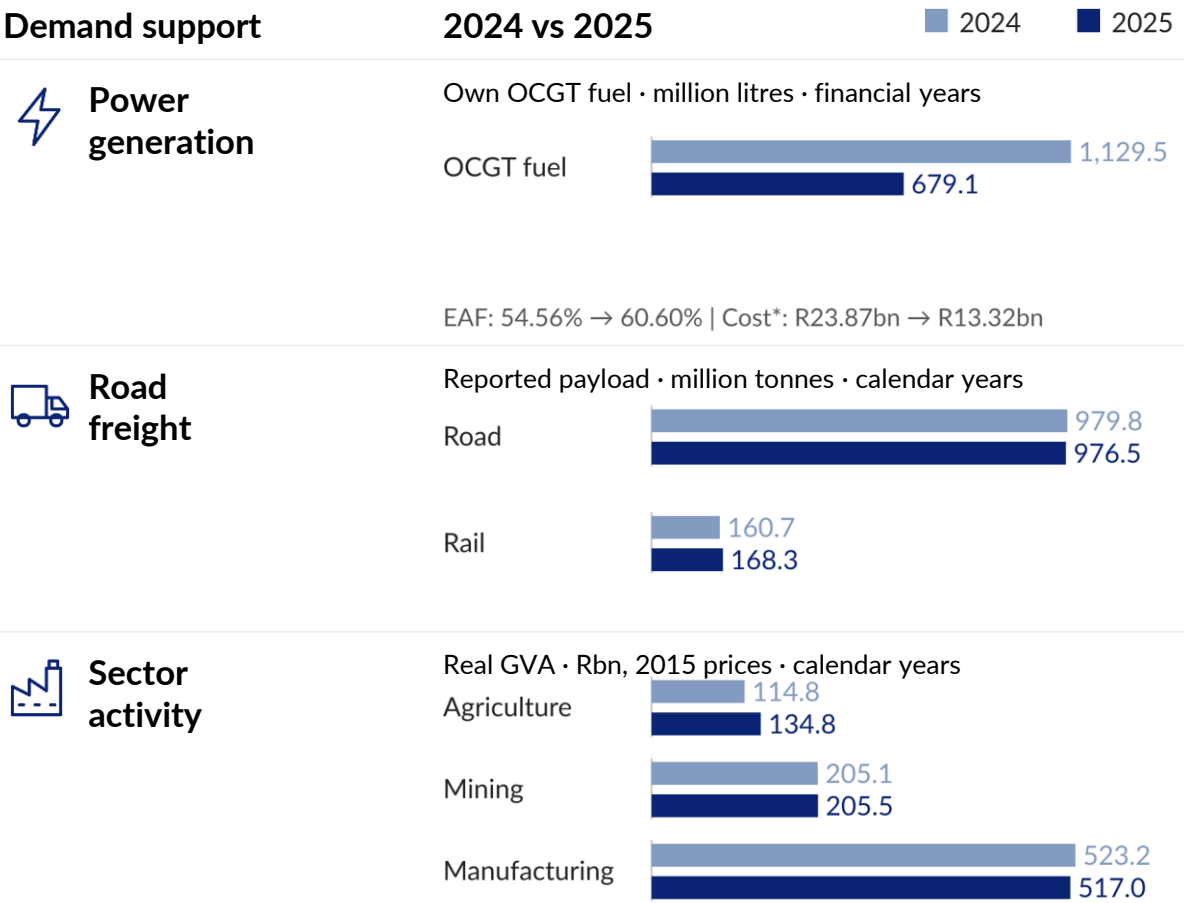
11.89 to 12.91 bn litres
+1.02 bn L / +8.6%
CAGR +0.8% per year

2023 Department sales total 21.94bn L. The 2024 FIASA comparison is 20.76bn L, unverified; no 2025 sales figure is available.

Department sales 2013-2023: petrol CAGR -2.08%, diesel +0.82%. FIASA 2024 is excluded from the trend. Workbook History Q71/R71. RAF levy evidence uses fiscal years and still needs reconciliation.

Power recovery reduces diesel support while freight and sector activity remain important

Demand support	2024 vs 2025	Implication for demand
Power generation	Own OCGT fuel · million litres · financial years OCGT fuel: 1,129.5 → 679.1 (-39.9%) EAF: 54.56% → 60.60% | Cost*: R23.87bn → R13.32bn	450m fewer litres in one year. Sustained recovery reduces diesel demand; retirements and replacement timing could reverse it.
Road freight	Reported payload · million tonnes · calendar years Road: 979.8 → 976.5 (-0.3%) Rail: 160.7 → 168.3 (+4.7%)	Road carries 85.3% of tonnes. Workbook truck-fuel proxy: 3.15bn L in 2023, partial operator coverage.
Sector activity	Real GVA · Rbn, 2015 prices · calendar years Agriculture: 114.8 → 134.8 (+17.4%) Mining: 205.1 → 205.5 (+0.2%) Manufacturing: 523.2 → 517.0 (-1.2%)	2025 activity-based diesel estimates: mining 1.406, industry residual 0.165, agriculture 1.131bn L. Uses can overlap.

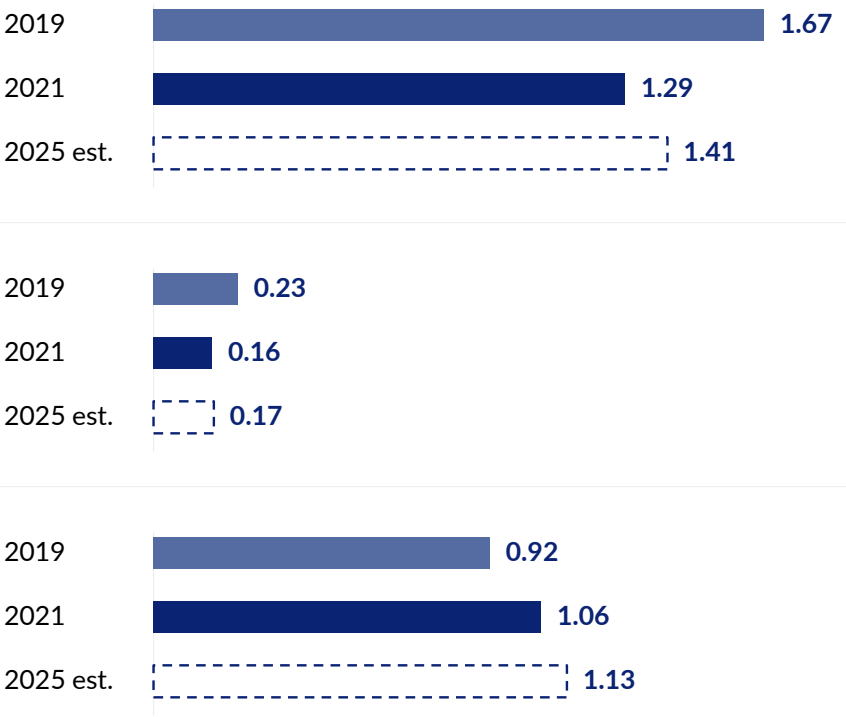


Eskom Integrated Report 2025 pp100,141: own-fleet diesel + kerosene; *cost includes storage/demurrage. Stats SA P7162 Dec 2025 p7: both years from the revised vintage. Stats SA P0441 Q2 2026 annual real GVA: agriculture includes forestry/fishing. FY and calendar-year periods shown separately. Bars start at zero; scales are common within each panel, not across different units. Light blue = 2024; navy = 2025. Workbook Sector history: post-2021 litres estimated. Freight proxy = survey fuel spending / wholesale price; partial coverage.

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Mining, manufacturing and agriculture link diesel demand to sector activity

Sector	Diesel use, billion litres / year	Activity and interpretation
Mining	2019: 1.67 2021: 1.29 2025 est.: 1.41	Mining activity drives the estimate. Average historical intensity and road-use overlap remain uncertain.
Manufacturing / non-specified industry	2019: 0.23 2021: 0.16 2025 est.: 0.17	Industry residual excludes mining and construction. This is an estimate, not a complete industrial census.
Agriculture / forestry	2019: 0.92 2021: 1.06 2025 est.: 1.13	2025 real GVA R134.8bn, 2015 prices. Weather, irrigation and transport affect fuel intensity.



Bars start at zero; all sectors use the same 0-2bn litre scale. Outlined bars are estimates.

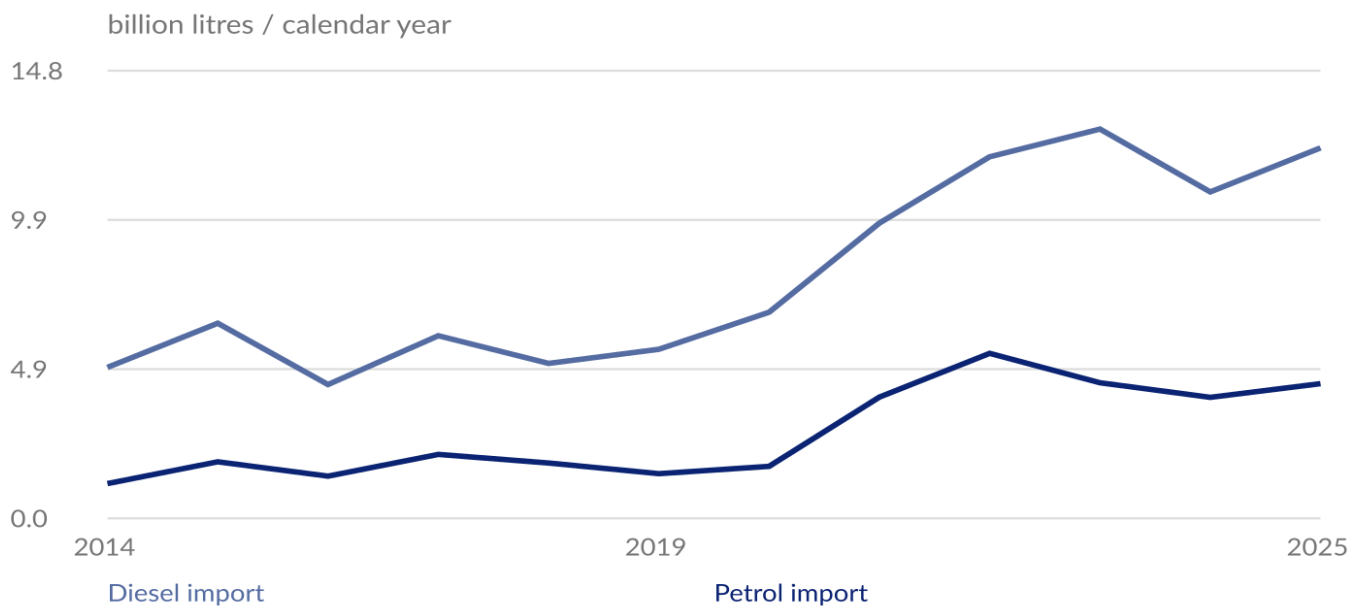
2025 estimates apply average 2018–2021 intensity to sector activity. They require reconciliation with road use and total diesel sales.

Manish master workbook, DR01 sector activity: 2025 activity × historical average intensity. Department energy balance ends in 2021. Later sector litres are estimates; attribution and road overlap remain open.

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Refinery shutdowns have increased reliance on imported petrol and diesel

Import trend



Change since 2020

01 | Diesel import

6.81 to 12.25 bn litres
Increase: 5.43 bn litres
CAGR: 12.4% (2020–2025)

02 | Petrol import

1.72 to 4.45 bn litres
Increase: 2.73 bn litres
CAGR: 21.0% (2020–2025)

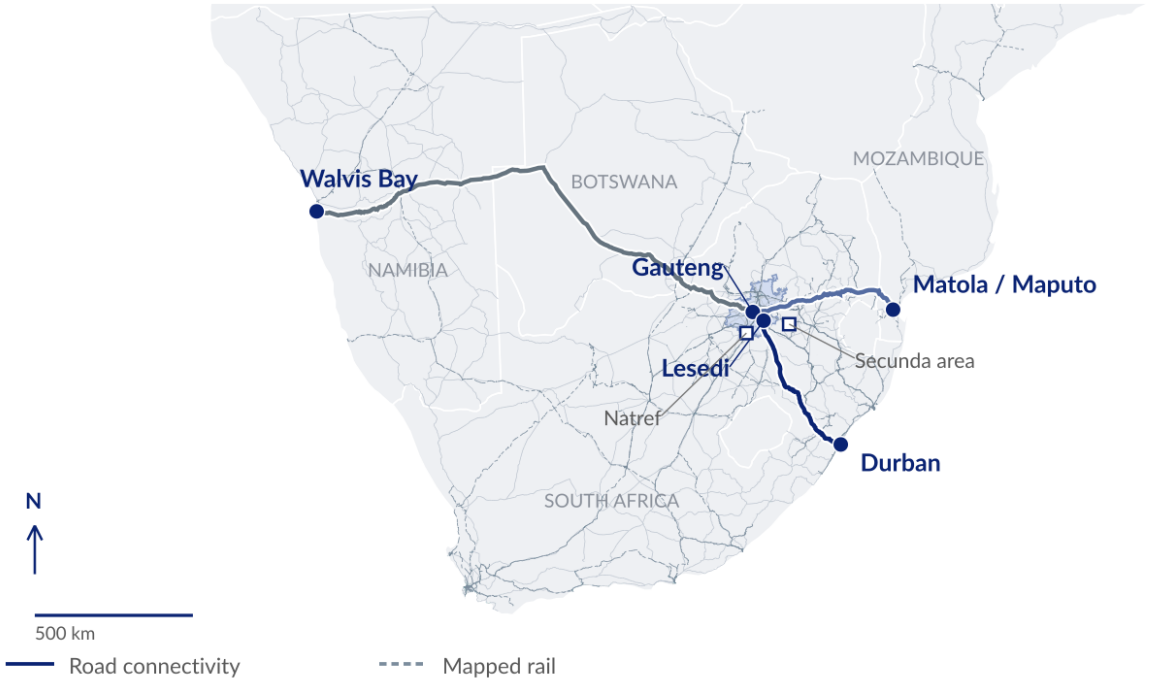
More inland handling opportunity; capture and turnover determine benefit.

2025 imports: diesel 12.25bn L and petrol 4.45bn L, net 15.15bn L. Durban customs accounts for 13.16bn L; terminal capture remains unverified.

SARS litre-reported petrol/diesel codes, complete calendar years. 2020–2025 changes are historical comparisons, not plant-level causal attribution. Export histories retained in the source report; unit/blend exclusions remain.

Emerging Maputo and Walvis Bay routes must be tested alongside Durban and domestic supply

Competing supply corridors



OSM geometry; routes test connectivity, not tanker access or operating service.

Same product, same destination

01 | Compare delivered cost

Product, handling, transport, border, losses and inventory finance.

02 | Establish price headroom

Bearable charge less actual logistics cost. Capacity and reliability constrain capture.

03 | Include domestic sources

Secunda and Natref also serve inland demand; routes and access need validation.

Next action

Match route costs and access to the same product and Gauteng destination.

MPP use averaged 105ML/week in FY2025 against 148ML/week capacity. Compare route costs and access by destination; average spare capacity gives no Vopak entitlement.

© OpenStreetMap contributors (ODbL; osm.org/copyright) / Geofabrik; Natural Earth; geoBoundaries. Shared LAEA projection. Paths test road connectivity, not tanker access. Secunda uses a city reference; Natref a mapped site. Terminal access, turn restrictions and operating rail/pipeline services remain unvalidated. Emerging alternatives are in scope; relative competitiveness is not yet quantified.

Durban and Lesedi connect coastal supply to Gauteng; operating data must establish room for growth

Asset / measure	Established position	What must be established
Durban	360,246 m³ published gross capacity. Coastal receipt and onward distribution; mixed products.	146,000 m³ estimated working subset from the 162,000 m³ expansion. Full petrol/diesel allocation still needs confirmation.
Lesedi	140,000 m3 clean-product capacity. Includes the 40,000 m3 expansion commissioned in October 2025.	124,000 m³ working estimate: 88,000 documented plus 36,000 estimated for newer tanks. Actual throughput remains open.
Operating constraints	MPP use/capacity: 105 / 148ML per week in FY2025. Shared annualised use 5.46bn L.	Measure receipt, pipeline and dispatch limits, seasonal peaks and stock requirements.
Implication for growth	Actual throughput and achievable turns are not yet established in this review.	No spare capacity is claimed. Test captured demand against verified handling limits before selecting investment.

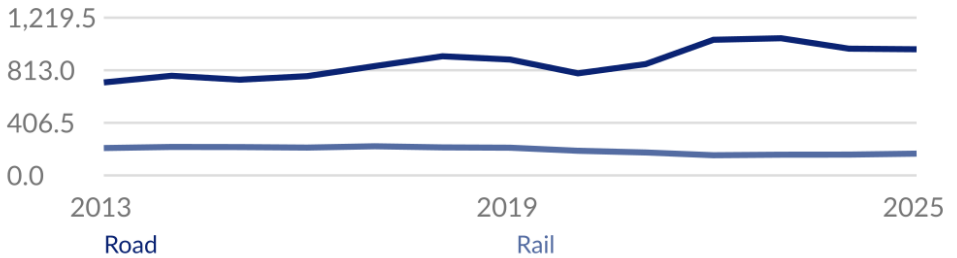
Vopak terminal directory; NERSA Lesedi licence documents; registered terminal_site_assumptions.csv; Transnet Pipelines 2025 p8. Gross capacity, estimated working tankage and annual throughput have different meanings.

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Rail recovery, EV adoption and combustion-vehicle efficiency could reduce transport fuel demand

Rail recovery

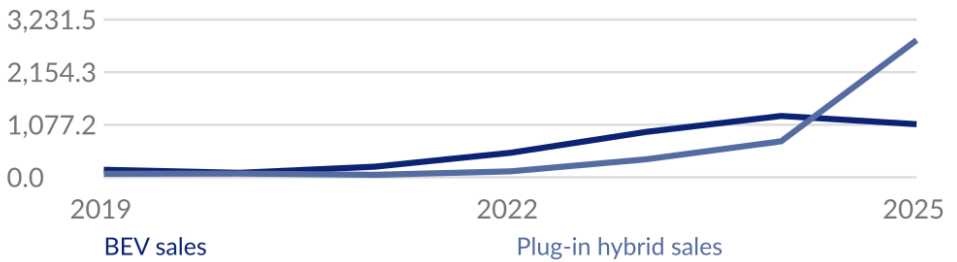
million tonnes



D2: capacity, reliability, transferable freight
Tonnes require haul distance and traction before conversion to net diesel savings.

EV uptake

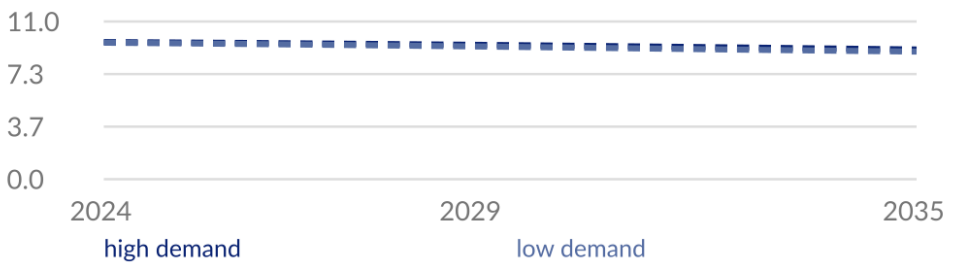
vehicles sold / year



D3: affordability, charging, fleet replacement
Sales history is not fleet penetration.
L/M/H adoption calibration remains open.

Fleet efficiency

L/100 km; draft engine



D4: new-cohort efficiency and mileage
Dashed paths are existing model diagnostics.
OEM history and Medium calibration are open.

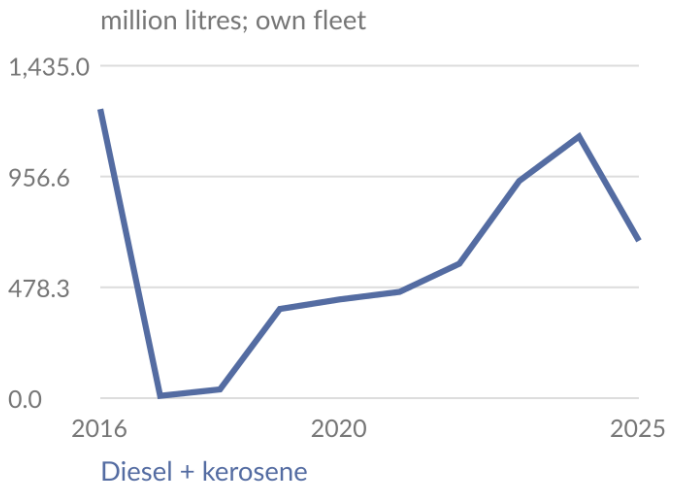
2025 BEV sales were 0.18% of new sales. Manish reviewed the implemented EV/efficiency pair; hybrid savings and rail diversion still need explicit combination rules.

Stats SA freight; naamsa all-market sales 2019-2025, Workbook Vehicle history S31:S36. BEV/PHEV/HEV sales are distinct from fleet shares. Efficiency paths remain model assumptions, with OEM history and annual calibration open.

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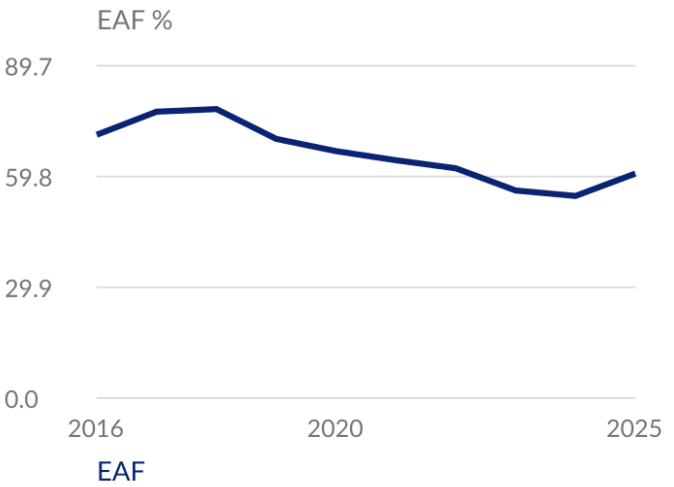
Power generation recovery could lower diesel use further, while outages and replacement delays could sustain domestic demand

Diesel use



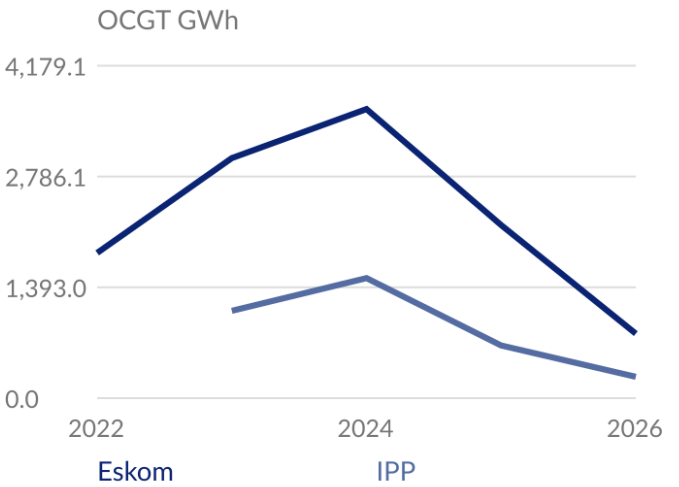
01 | Separate dispatch demand
Separate Eskom from private backup. Load-shedding and backup histories are still needed.

Fleet availability



02 | Test recovery durability
Test outages and retirement timing. Sustained availability reduces pressure on diesel generation.

Peaking generation



03 | Time replacement generation
Wind, solar and gas timing matters. Calibrate dependable generation before forecasting diesel litres.

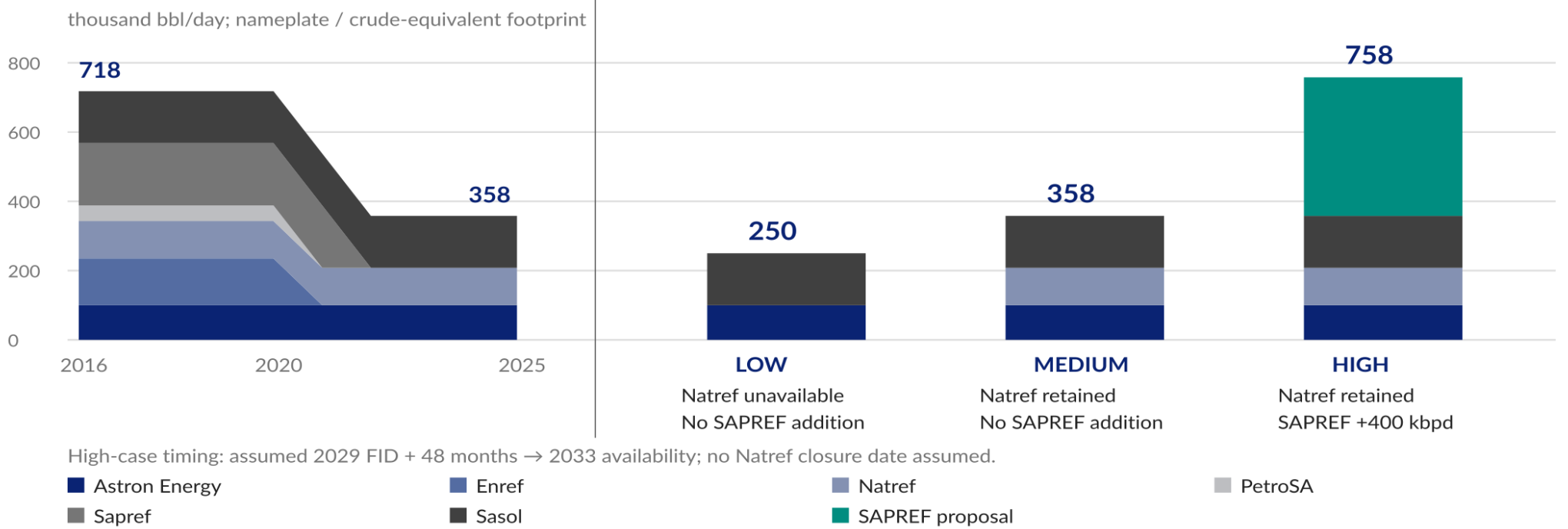
The engine still includes 3,000MW of notional plants. Reconcile power and road baselines jointly against reported Eskom fuel and estimated IPP burn before forecasting.

Eskom FY ending March: reported own diesel + kerosene; Eskom/IPP GWh include kerosene output. Workbook Power fleet separates the four diesel stations. Gas switch, coal-site repowering and L/M/H settings await Nigel/Henry review.

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SAPREF could expand the supply footprint, while Natref unavailability would reduce it

Reported history | **Illustrative 2036 capacity cases**



SAPREF's import-terminal phase can compete before any refinery restart. Natref availability and Secunda output also matter; these capacity cases are not product forecasts.

FIASA 2025 p49; registered review_capacity_scenarios.yaml and review_supply_worlds.yaml. Illustrative 2036 sensitivities, not calibrated production forecasts. Other plants held at 2025 reported values; Secunda gas/MRG and PetroSA restart effects remain unquantified. Low is not a Natref closure prediction.

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Emerging gateways could redirect inland growth; delivered cost and reliable access determine capture

Competing route	What changes the choice	Investment implication
Durban–Lesedi	FY2025 MPP 105 / 148ML per week. Match tariffs, injection rights and reliable inland dispatch by customer.	Test access and handling improvements that protect competitive customer supply.
Maputo / Matola	Delivered cost, border time and capacity into each inland destination.	Include as an emerging alternative; test customer exposure and corridor responses.
Walvis Bay	Distance, transit, border costs and feasible destination coverage.	Consider emerging inland opportunities; compare reach customer by customer.
Domestic sources	Secunda / Natref availability, product output and distribution access.	Test inland handling and connections as well as import infrastructure.

Compare common product, destination, price date and tax basis. Emerging gateways must be considered; no relative cost ranking is asserted without quotes and verified access. Landlocked SADC opportunities require separate market/access evidence.

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Choose the investment that removes the proven constraint—and earns a return

Investment option	When it becomes relevant	Decision / commitment trigger
Existing assets / debottlenecking	Customer growth can be served through better tank turns, scheduling or receipt/dispatch.	Use available working capacity; invest in the specific bottleneck where incremental value is positive.
Corridor / rail access	Competitive customer flows are constrained by transport capacity, cost or reliability.	Secure service and volume commitments. Compare access contracts, loading assets and partnership investment.
Additional tank capacity	Profitable captured flows exceed achievable handling and storage remains the binding constraint.	Size and stage tanks against contracted demand; pass the four tests and downside cases.
EV-related adjacencies	Identified charging / fleet-energy customers offer a viable opportunity as transport electrifies.	Screen partnerships or dedicated assets separately: demand, power access, strategic fit and return must be proven.

Investment menu, not approved projects. Network rail recovery is an external fuel-demand driver; Vopak rail loading/access is a distinct investment option. EV uptake reduces some fuel demand; it does not itself prove an EV investment case. Project costs, contracts and returns remain open. Three fuel-logistics packages are priced illustratively on p40; these do not constitute engineered rail or EV project estimates.

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Resolve customer volume, working capacity and project economics first to determine the investment

Outstanding request / coordinator	Required data / handback	Decision supported
DR01 | Manish; Nigel review	Received: sales, customs and energy balance to 2021. Open: stocks, product output after 2021 and comparable 2024–25 sales.	Reconciled current market scale.
DR04 | Manish; Nigel review	Received: entry, tariff and FY2025 MPP evidence. Open: customer access, route quotes and modal flows.	Accessible Durban and Lesedi markets.
DR07/08 | Manish; Nigel/Henry review	Received: power, fleet, refinery and EV/efficiency review. Open: joint baseline and hybrid/rail rules.	Demand and domestic-output paths.
DR02/05 | Nigel coordinates	36 months of site/product movements, shared transfers, destinations, customer commitments and contracts.	Actual market share and achievable capture.
DR03 | Nigel coordinates	Working/committed tankage, turns, occupancy, downtime and receipt/dispatch limits.	Operating headroom and bottlenecks.
DR06 | Nigel coordinates	Storage/handling terms, costs, capex, timing, tax, working capital and hurdle rate.	Incremental returns and investment scope.

Public handback DR01/04/07/08 received and updated through 9 October. Client DR02/05/03/06 prepared; sending and receipt not recorded. Review workbook covers all nine requests, including separate DR09 EV adjacency.

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Appendix

Supporting evidence and calibration

Durban–Lesedi corridor into Gauteng

10 October 2026

Detailed evidence, assumptions and remaining calibration requirements

Photo source: Royal Vopak | Terminal Durban | vopak.com



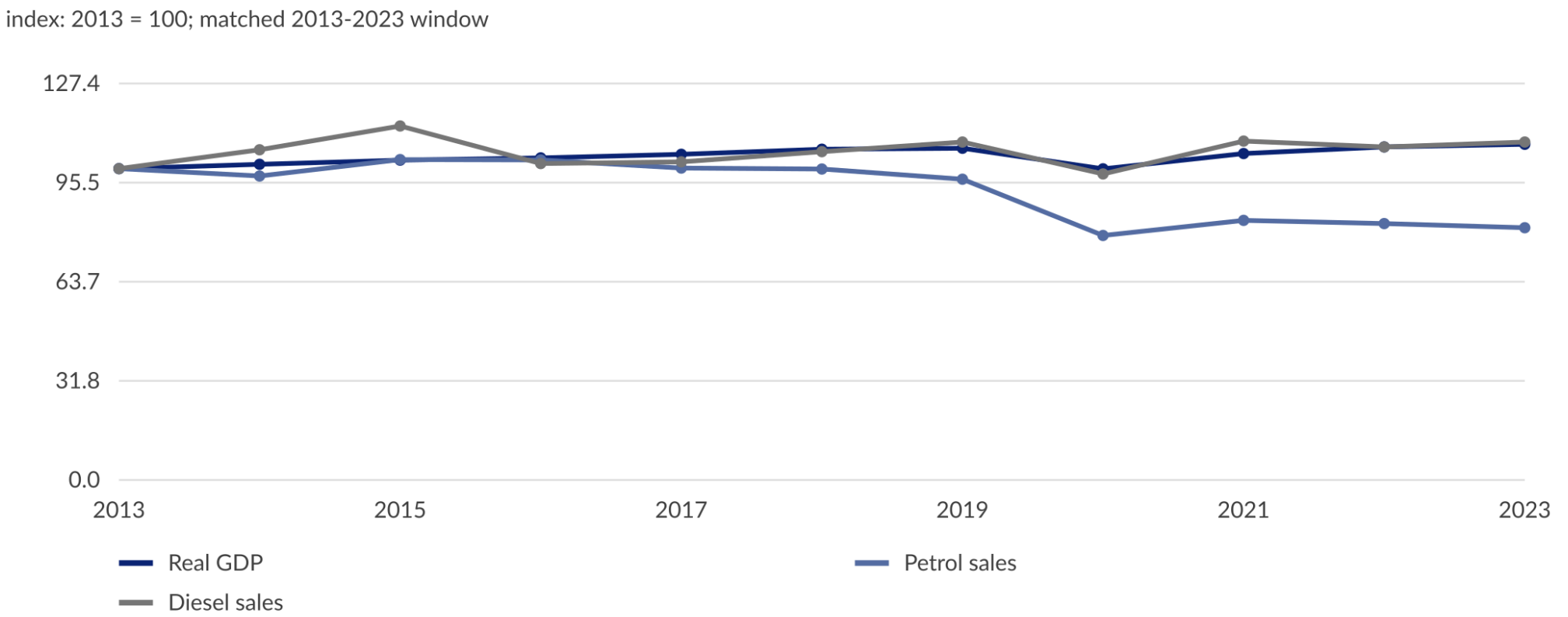
The supporting evidence calibrates demand, supply and the terminal investment tests

Evidence group	Detail retained	Connection to the story
Demand	Workbook sector history, vehicle stock/sales, diesel-use estimates, freight and power fleet.	Observed history and labelled estimates inform D1-D5; annual L/M/H paths still require agreement.
Supply	Plant output, gas availability and restart timing	S1–S4 create the three domestic-supply paths.
Capture and operations	Routes, customer share, working tankage and turnover	T1–T5 translate each world into terminal flows and constraints.
Returns	Rates, costs, capex and funding; separate EV adjacency economics.	F1 translates each investment option into cash flow and returns.

The main story runs from the opening thesis to the investment tests. The added lever register and power/supply calibration pages record what remains to be populated.

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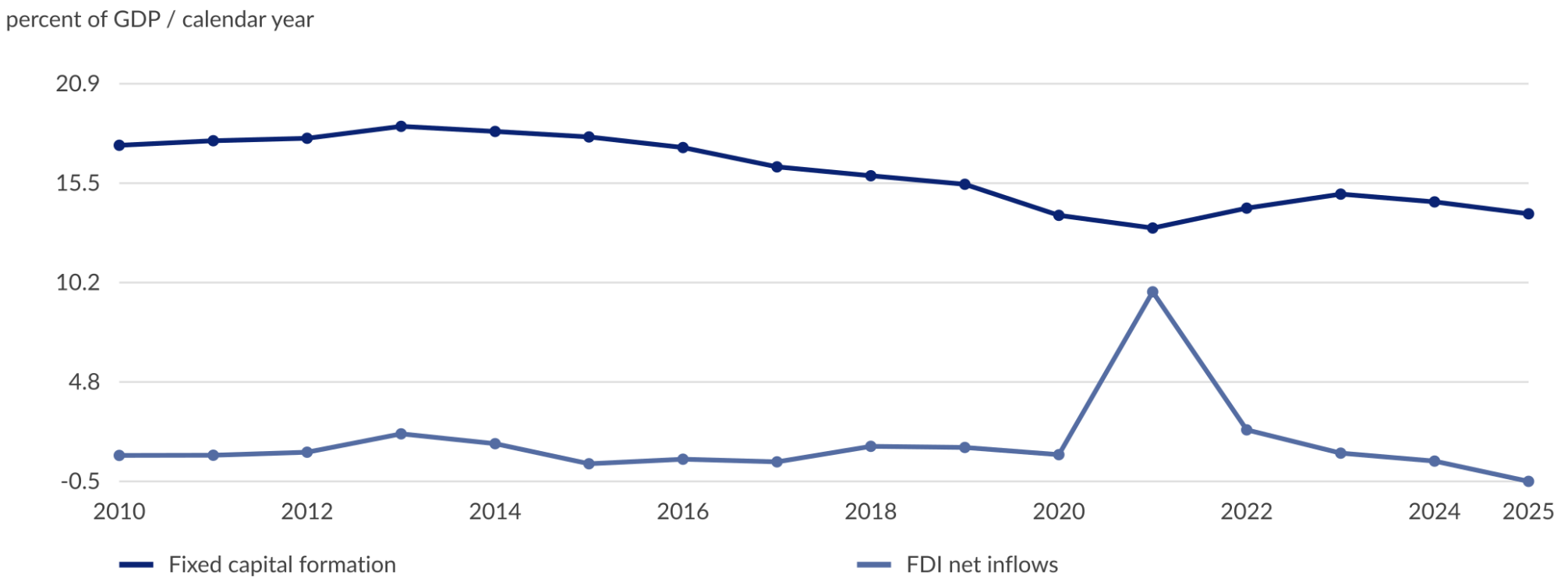
Fuel demand has diverged from GDP, so sector activity and fuel intensity need separate treatment



The workbook separates mining/manufacturing indices and agricultural GVA from fuel intensity. Post-2021 sector litres are estimates; GDP alone cannot explain fuel demand.

Staged GDP and department fuel sales, indices 2013=100. Sector attribution remains open; COVID distorts short-period comparisons.

Fixed investment has weakened while FDI fluctuates, limiting a simple growth-led fuel argument



Fixed investment helps frame productive capacity. FDI can include large transactions; neither series is a direct investor-confidence measure or an additional fuel-demand multiplier.

World Bank WDI, FDI net inflows and gross fixed capital formation as % of GDP, 2010–2025. Confidence series remains an evidence gap.

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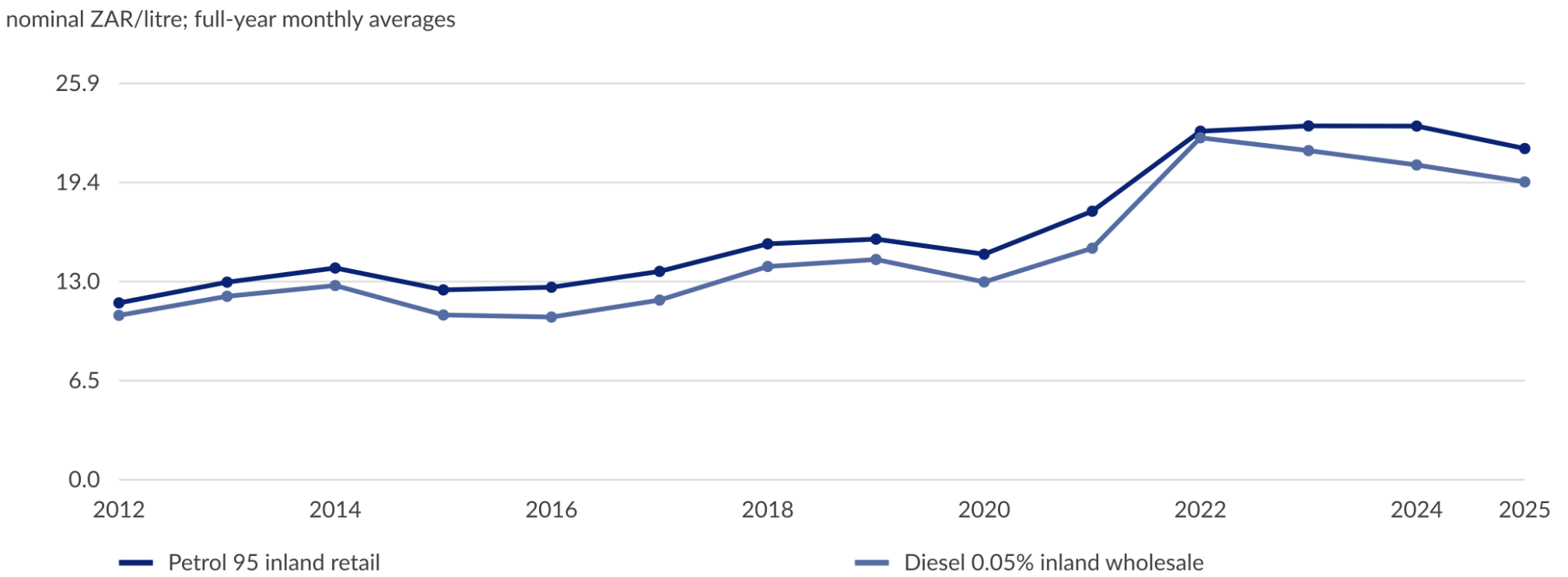
Mining, agriculture and aviation expose fuel demand to different activity drivers

Evidence	Interpretation	Implication for Vopak
Mining	2025 activity-based diesel estimate: 1.406bn L. Post-2021 fuel volumes are estimated.	Use activity and intensity separately; balance intensity varies materially.
Manufacturing	2025 industry-residual diesel estimate: 0.165bn L. Sector boundary needs review.	Industry less mining/construction. This is not a complete industrial diesel census.
Agriculture	2025 real GVA R134.8bn (2015 prices); indicative diesel 1.13bn L.	Fuel estimate holds 2018-2021 intensity; crop and irrigation exposure remain open.
Aviation	2024 jet sales 1.96bn L; 456,214 ACSA movements; 4,285 L/movement.	FIASA sales unverified; movements omit non-ACSA airports. Fuel mix and routes are not separated.

Workbook Sector history R7/R9/R17/R19/S23/S25 and History R117/R124/R125. Sector litres after 2021 are estimates; jet 2024 uses unverified FIASA. Index base 2019=100; GVA in 2015 prices.

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Higher fuel prices strengthen EV running-cost incentives, but ownership economics determine uptake



Shipping disruption, crude prices and FX can move fuel costs temporarily. EV adoption depends on sustained total ownership cost, purchase affordability, charging access and fleet replacement.

Department fuel-price history, nominal R/litre. No quantified geopolitical premium or calibrated EV adoption response is asserted.

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New-vehicle efficiency gains affect fuel demand gradually as older vehicles leave the fleet

Evidence	Interpretation	Implication for Vopak
Fleet growth	2025 cars 8.18m; LCVs 2.78m; trucks 0.405m.	Registered stock shows scale; fuel within class and opening cohort age remain unknown.
Replacement	2025 apparent retirement: cars 2.36%, LCVs 3.42%. Model settings: 4% and 5%.	Stock plus sales less next stock is a diagnostic, including deregistration, exports and classification effects.
Efficiency / EVs	Study mileage/fuel-use benchmarks and separate BEV, PHEV and hybrid sales are available.	Implemented EV/efficiency pair reviewed. New-cohort efficiency applies to combustion vehicles. Avoid counting hybrid savings twice.

Workbook Vehicle history S6:S8/S24/S26. Apparent retirements are not verified scrappage. Stone et al. study uses a 2014 base; no recent OEM efficiency history or fuel-by-class stock is established.

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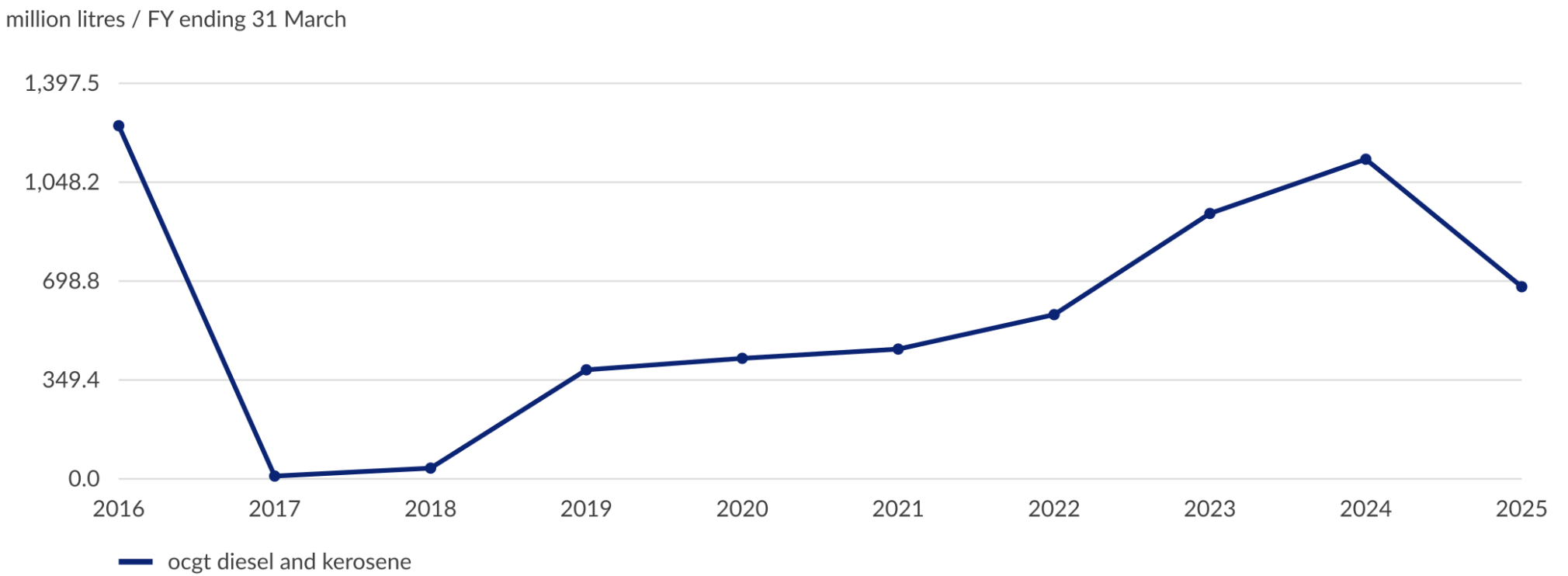
Vehicle stock and sales expose fuel-demand calibration gaps

Calibration item	Observed evidence	Model / calibration reference	Implication / limitation
2025 registered fleet	Cars 8.18m; LCVs 2.78m; trucks 0.405m.	Opening cohorts need review.	Age and fuel within class are missing; all-class fuel counts are available only for December 2023.
New-sales mix, 2025	Cars 70.7%; LCVs 24.1%; medium/heavy 5.2%.	78% / 18% / 4%.	Observed class mix differs. Agree segmentation before adopting a replacement.
Apparent retirement, 2025	Cars 2.36%; LCVs 3.42%.	4% cars; 5% LCVs.	Derived from stock and new sales; includes classification changes, exports and deregistrations.
New drivetrains, 2025	BEV 1,088; PHEV 2,810; conventional hybrid 12,818.	BEV observed share 0.18%.	Fuel/class splits and electric-LCV paths remain estimates. Sales shares do not establish fleet shares or hybrid fuel savings.
Petrol-use diagnostic, 2023	1,056 L per registered petrol vehicle.	Passenger setting implies 1,615 L/year.	Aggregate versus passenger scope differs. Investigate mileage and fuel-use calibration, not a direct replacement.

Workbook Vehicle history S6:S8/S18:S20/S24/S26/S31:S36/Q52. NaTIS and naamsa classes are not identical. Apparent retirement is not verified scrappage; petrol diagnostic compares different scopes.

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Eskom’s own OCGT fuel use fell 40% in FY2025 as availability improved

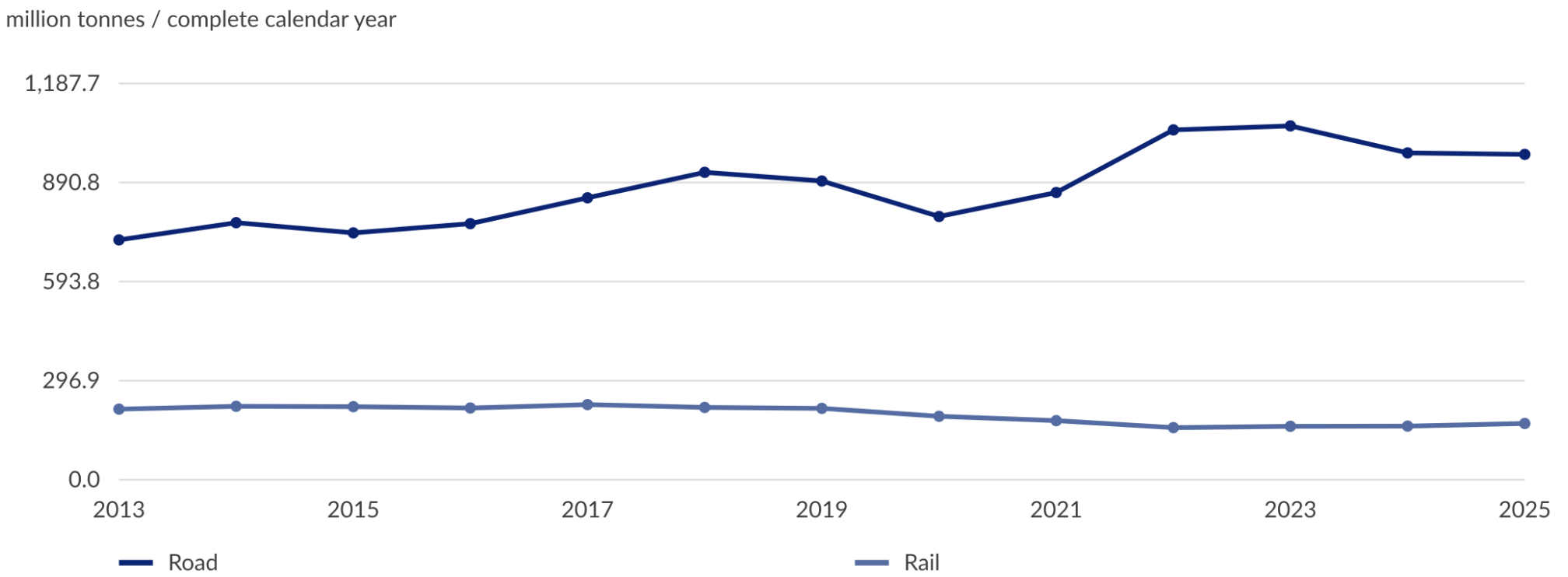


Own-fleet fuel fell from 1,129.5 to 679.1m L. FY2025 Eskom plus estimated IPP fuel is ~0.89bn L; the model’s power/road split needs joint recalibration.

Eskom Integrated Report 2025, FY March, own diesel + kerosene. Workbook combined generation includes IPPs and kerosene stations; multiplying it by 0.31 L/kWh is an estimate, not reported diesel burn. Private backup remains unmeasured.

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Road carries most reported freight tonnage, making transferable haul distance a key diesel lever



Apply freight diversion once to a defined road baseline. The ~3.15bn L operator-spending proxy has partial coverage and does not establish corridor tonne-km.

Stats SA payload and transport-industry fuel purchases. Workbook Diesel by use Q28: spending / average diesel price, hire-and-reward firms only; may include petrol/lubricants. Road residual includes unseparated uses; tonne-km and rail traction remain open.

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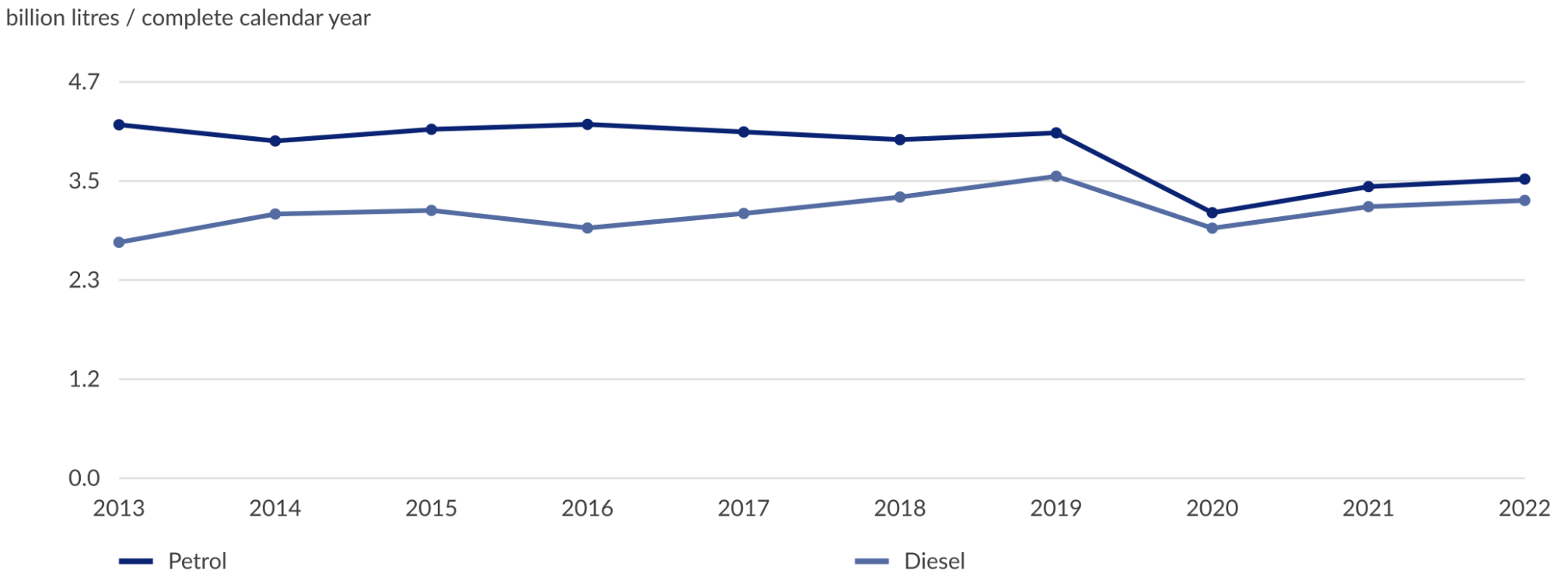
Rail funding and reliable service determine how much road freight can return to rail

Evidence	Interpretation	Implication for Vopak
Network ambition	The supplied brief reports a 250 Mt target dependent on a funding plan.	Do not treat every additional rail tonne as road displacement. Commodity, corridor and origin matter.
Funding constraint	Brief: R46.34bn approved over five years, including R1.71bn expansion.	Maintenance and sustaining spend dominate. A published plan is insufficient evidence of delivered capacity.
Container corridor	Brief: R2.11bn in 2026/27–2028/29; private funding expected thereafter.	Prioritise Durban–Gauteng road/rail competition and reconcile the corridor to Network Statement v4.
Two fuel effects	Rail can displace truck diesel while improved infrastructure also changes fuel-distribution economics.	Apply road-to-rail diversion once to eligible freight activity. Separately assess the route used to distribute fuel inland.

User-supplied AIA Transnet Funding Brief, Sep 2026, pp1–2; cites FY2026 AFS and Network Statement v4 Annexure 17. Primary figures not independently reconciled.

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Gauteng demand establishes market scale, while Lesedi capture and turnover determine throughput



Gauteng sales were 6.81bn L in 2022; the six inland provinces totalled 11.95bn L. These are nested catchments, not low/high Lesedi market scenarios.

Department provincial sales through 2022; workbook History Q84/Q104 and R84/R104 estimate later volumes. 2024 also uses unverified FIASA national sales. The chart retains observed 2013-2022 sales history, not current market share.

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Every market world passes through explicit demand, supply, capture and operating levers

Lever IDs	Inputs to calibrate	Output / remaining evidence
D1–D5 demand	D1 sector growth; D2 rail; D3 EV adoption; D4 efficiency; D5 power.	EV/efficiency pair reviewed. Hybrid, freight/rail and baseline combination rules remain open. Proposed annual paths require agreement.
S1–S4 supply	S1 SAPREF; S2 Secunda gas/MRG; S3 Natref; S4 PetroSA.	Output, yields and feedstock dependencies remain open. Reconcile proposed capacity, utilisation and gas assumptions.
T1–T2 capture	T1 delivered route economics/access; T2 customer/product/destination capture.	Allocate domestic and imported fuel to unique terminal deliveries. Tariffs, rights and commercial allocation remain open.
T3–T5 operations	T3 eligible working tankage; T4 turnover; T5 receipt/dispatch, peaks and stock policy.	Working-tank sensitivity uses Durban 146,000 and Lesedi 124,000 m^3, both partly estimated. Validate turns, peaks and route access.
F1 finance	Storage/handling terms, costs, capex, tax, working capital and hurdle rate.	Translate constrained throughput into cash flow, NPV, IRR and investment triggers. Commercial inputs remain open.

Workbook HML response has 29 levers. Examples: BEV adoption, rail diversion, mileage, prices, hybrids, truck/LCV electrification and neighbouring exports. All replacement ranges remain proposals; no forecast inputs are adopted in this pack.

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Generation timing must close the power balance before residual diesel can be forecast

Time series	Assumption / lever	Evidence status and next use
Recovery history	Own Eskom fuel/EAF and Eskom/IPP GWh are assembled; generation-based litres are estimated.	Align FY/calendar scope. Reconcile the model's 3.58bn L power estimate with reported/estimated fuel and other demand uses.
Fleet and retirements	Four diesel stations: 3,089 MW. Avon/Dedisa PPA endings and coal retirement dates are assembled.	Contract end is not confirmed shutdown. Siting, renewal and actual coal decisions remain open.
Replacement / gas	Ankerlig/Gourikwa gas switch and coal-site turbines are case dependencies, not committed outcomes.	Engine includes 3,000MW of notional plants. Treat these separately from the four existing stations and proposed coal-site cases.
Proposed L/M/H cases	2030 power diesel: 86 / 637 / 4,224m L in the workbook scenarios.	Nigel/Henry review D22: conversion rate, backup share, load factors, repowering location and gas supply.

Workbook Power fleet F11:S15 and N63:N65; proposed scenarios, not forecasts. Generation includes kerosene. IPP renewal, gas availability and coal-site repowering are not established. Historical burn must match scope and period.

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The diesel fleet and replacement timing define the power-demand cases

OPERATING DIESEL FLEET: 3,089 MW

Diesel station	MW	Owner
Ankerlig	1,338	Eskom
Gourikwa	746	Eskom
Avon	670	Independent
Dedisa	335	Independent

TIMING AND DEPENDENCIES

Avon / Dedisa PPAs end in 2031 / 2030.
Five coal sites have exemptions to March 2030; final outcomes remain open.
Gas-switch and turbine locations are assumptions, not committed projects.

PROPOSED WORKBOOK CASES — NIGEL / HENRY REVIEW

Proposed case	2030 m L	2035 m L	Key dependency
Low	86	20	Gas switch; PPAs end; no coal-site turbines.
Medium	637	637	3 GW at coal sites; gas available; diesel backup.
High	4,224	4,224	6 GW at coal sites; no gas; diesel peakers.

The engine also contains 3,000MW of notional capacity. Reconcile it with these cases and historical power/road use before adopting a forecast.

Workbook Power fleet F11:F15, N63:N65 and S63:S65. Proposed cases, Nigel/Henry review D22. 0.31 L/kWh, backup fraction, load factors, gas supply and coal-site repowering need review. Kerosene stations and private backup are separate.

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SAPREF and Natref define capacity sensitivities; plant output must complete the supply paths

Plant lever	Required time-series inputs	Scenario treatment
S1 SAPREF	High capacity illustration adds 400 kbpd from assumed 2033 availability.	SAPREF plans an import-terminal phase before refining. No committed refinery restart date or product slate is established.
S3 Natref	108 kbpd remains in Medium/High; excluded in the Low 2036 stress case.	Natref is separate from Secunda. FY2026 Sasol output uses co-owner capacity; do not scale by historic ownership mechanically.
S2 Secunda	All-refined-products output: 29.1m barrels FY2024, 27.6m FY2025 and 30.6m FY2026.	June fiscal years; product mix differs. Gas/MRG affects liquid output; do not equate barrels with diesel litres.
Other plant inputs	Astron output evidence is available; its petrol/diesel 50/50 split is an assumption. PetroSA restart remains uncertain.	Confirm product yields and actual operator output. No PetroSA addition appears in the existing capacity illustration.

Manish DR08 refinery evidence, updated 9 October; Sasol FY2026 metrics; Glencore/Astron evidence; CEF SAPREF roadmap. FY output, calendar capacity and product yields differ. Capacity sensitivities remain illustrative.

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A conditional Durban import-capture screen gives a 2.63bn litre base case

Illustrative case	Annual physical volumes	Assumptions and interpretation
Low **10% gateway share**	Durban 1.32bn L Lesedi 0.66bn L Unique flow 1.32bn L	50% of captured Durban flow routed to Lesedi. Conditional on customer allocation and terminal/pipeline access.
Base **20% gateway share**	Durban 2.63bn L Lesedi 1.32bn L Unique flow 2.63bn L	50% of captured Durban flow routed to Lesedi. Conditional on customer allocation and terminal/pipeline access.
High **30% gateway share**	Durban 3.50bn L Lesedi 1.75bn L Unique flow 3.50bn L	50% of captured Durban flow routed to Lesedi. Durban tank screen caps the requested 3.95bn L at 3.50bn L.

10/20/30% shares and 50% routing are authored sensitivities. Base turns fixed at two/month. Lesedi is a subset, so volumes are counted once. Domestic and other-terminal Lesedi receipts are excluded.

Registered market_review.yaml and terminal_site_assumptions.csv; 2025 Durban customs proxy 13.1605bn L. No forecast probabilities, measured market share or incremental revenue. Product mix, access and working tankage need validation.

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A matched demand snapshot is available, but the domestic supply balance still needs reconciliation

Evidence / bn litres	Petrol	Diesel
2023 Department sales	9.04	12.91
2024 FIASA sales, unverified	9.03	11.73
2025 SARS imports / exports	4.45 / 0.80	12.25 / 0.75
2025 sales / product output	Sales absent; product production series ends 2021.	Sales absent; product production series ends 2021.
RAF levy cross-check	Combined levied volume 24.42bn L, year to March 2025.	Fiscal/accrual versus calendar sales. Reconcile coverage before interpreting the difference.
Decision consequence	2025 gross imports 16.70bn L; net 15.15bn L. Durban customs 13.16bn L.	Current import penetration cannot be calculated using 2023 sales. Post-2021 production and stocks remain open.

Workbook History Q59/Q64/R59/R64/S36:S40/R70. RAF-derived 24.42bn L is combined fuel, fiscal/accrual basis, not an adopted calendar sales total. Product output after 2021 and stocks remain missing; 2025 trade is not 2025 consumption.

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The six-step bridge is implemented for screening; calibration now depends on the requested inputs

Work package	Implemented now	Input needed to complete the decision
1 | Baseline	Public handback received through 9 October. Review workbook maps all nine requests and source-level limitations.	Agree baseline year/source; reconcile production, stocks, scopes and fiscal/calendar periods.
2 | Route competition	Delivered cost and bearable-charge calculation; route readiness register.	Matched quotes, service capacity and access.
3 | Vopak capture	Conditional Durban import-capture cases now quantified. Shared Lesedi transfers count once, see p36.	Customer allocation and unique terminal deliveries.
4 | Existing assets	Required-turn and operating-bottleneck calculation; gross-turn screen.	Working capacity, achievable turns and peak limits.
5 | Investment options	Pre-tax break-even volume calculation.	Commercial inputs and full incremental cash flows.
6 | Nine worlds	Nine-cell output register and demand/supply balance calculation.	EV/efficiency implementation review complete for the pair. Baseline, hybrid/rail interactions and annual paths remain open.

Screening functions tested with synthetic cases; commercial outputs remain unpopulated. Not integrated into the forecast engine or independently reviewed. Owners, request IDs and expiry triggers: workstreams/WS0_governance/workplan/sa_investment_bridge_2026_10_08.csv.

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Demand, supply and corridor access translate into captured throughput and required tank turns

Argument	Situation	Complication	Resolution / decision
Transport demand	2023 petrol/diesel sales total 21.94bn L; sector activity and fleet evidence explain different fuel exposures.	Sector growth, rail recovery, EVs and efficiency change fuel demand.	D1–D4 define activity, rail, EV and efficiency paths. Translate them into capturable product demand.
Power demand	Power shortages supported diesel use; Eskom consumption has recently fallen.	Recovery may persist or reverse through outages and replacement delays.	D5 adds fleet recovery, replacement timing and diesel dispatch to the same demand paths.
Domestic supply	Reduced local production has increased reliance on imports.	SAPREF, Natref and Secunda can change domestic fuel output materially.	S1–S4 define plant output and sourcing mix. Imports are one consequence of the domestic balance.
Corridor competition	Durban connects imported fuel to inland customers.	Alternative gateways, delivered costs, reliability and access alter flows.	T1–T2 compare delivered economics and capture for domestic and imported fuel serving common destinations.
Vopak investment	Durban and Lesedi provide connected assets whose throughput can rise through higher turnover.	Working tankage, receipt/dispatch, peaks and stock requirements constrain achievable turnover.	T3–T5 compare working capacity, required/achievable turns and dispatch limits. F1 tests incremental returns.

Lever IDs are mapped in the appendix. All nine market worlds use explicit routing, capture and operational assumptions. Tank occupancy, turnover and throughput utilisation are separate measures. Competitiveness varies by destination; neither alternative gateway is yet established as competitive into Gauteng.

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The revised illustration gives extra tanks R58m more NPV than improvement alone

High demand / low supply	Capital and throughput	Incremental value versus existing
Existing	R0m capex 1.860bn L/year served	NPV R+0.0m Unserved 0.953bn L/year
Improve access / dispatch	R100m capex 2.232bn L/year served	NPV R+24.2m Unserved 0.581bn L/year
Improve + add tanks	R600m capex 2.813bn L/year served	NPV R+82.2m Unserved 0.000bn L/year
Decision threshold	Improvement beats existing above 2.181bn L/year.	Tanks beat improvement above 2.765bn L/year; assumed capture 2.813bn.

ILLUSTRATIVE, 100% site basis, pre-tax at 10% for 20 years. Working tankage reduced from 126,000 to 124,000 m³. Model allocates more incremental contracted storage to added tanks, increasing assumed rent. This is model sensitivity, not new commercial evidence.

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Vopak reports anchor capacity and the revenue model; they do not disclose Lesedi project economics

Evidence class	What the sources establish	How it is used
REPORTED | Assets	Lesedi: 140,000 m^3; Durban: 360,246 m^3. Vopak share: 70% at each site.	Full physical capacity for operations. The commissioned 40,000 m3 Lesedi expansion is included in 140,000 m3.
REPORTED | Revenue	Storage services: EUR1,058.8m of EUR1,298.9m consolidated revenue in 2025.	81.5% is storage revenue. Rental may include minimum throughput; extra turns are not automatically extra revenue.
INFERRED | Group ratio	EUR93.9 revenue / gross m^3/year: proportional revenue divided by year-end proportional capacity.	Mixed-portfolio scale indicator only. Not a Lesedi tariff, margin or estimate of site revenue.
REPORTED | Benchmarks	15.6% group operating cash return; 5–7x capex/EBITDA for gas/industrial and 4–8x for transition infrastructure.	Context, not South Africa fuel-project hurdle rates. Authored 10% discount rate is separate.

Vopak Annual Report 2025 pp36/39, 63, 213 and 324; FY2025 Analyst Presentation p24; official terminal directory checked 8 Oct 2026. All other business units includes multiple countries; do not allocate its profit to South Africa. URLs, downloaded originals and hashes retained.

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The investment result rests on explicit assumptions that must be replaced or agreed

ILLUSTRATIVE input	Value used	What must be validated
Catchment balance	Demand 6.0 / 6.8 / 7.6bn L; domestic deliveries 2.4 / 3.6 / 4.8bn L.	Steady-state stress points; M rounds 2022 Gauteng sales for scale. No national production forecast implied.
Capture	70% of imports via Durban; Vopak captures 80%; 90% goes to Lesedi; 8% domestic capture.	Separate investment illustration with imported and domestic capture. Its shares differ from the Durban-only capture screen on p36.
Operations	124,000 m^3 working, 88.57% of gross; turns 1.25 existing / 1.50 improved; +40,000 gross m^3.	159,429 working m^3 with tanks at the same assumed fraction. Receipt/dispatch ceiling 3.5bn L/year remains illustrative.
Revenue	R1,500 / gross m^3/year rent on added contracted tanks; R60 / extra m^3 contribution.	Chargeable excess services only, excluding rent/minimum-throughput inclusions. Added rent follows incremental storage use.
Capital and costs	R100m improvement + R500m tanks; fixed opex R5m + R12m/year; maintenance 1% capex/year.	Author-selected cost sensitivities; no Vopak quotation or engineering estimate.
Cash flow	20 years at 10%; 50% / 75% / 100% operating ramp; capex upfront.	Pre-tax unlevered constant ZAR. Tax, working capital, construction timing and residual value excluded.

Every illustrative value is registered in assumptions/2026/vopak_investment_illustration.yaml. Not a forecast, commercial offer or approval. Proposed validation: Nigel coordinates Vopak commercial/operations; Manish reconciles; independent review outstanding.

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Capture and commercial assumptions can reverse the illustrated expansion choice

Independent sensitivity	Improve NPV Rm	Tanks NPV Rm	Preferred package
Base illustration	+24.2	+82.2	Improve + add tanks
Capex +25%	-2.9	-80.6	Existing
No chargeable excess services	-151.1	-366.7	Existing
Vopak gateway capture 60%	-10.9	-655.6	Existing
Captured volume -10%	+24.2	-258.1	Improve access / dispatch

The lower working-capacity estimate increases assumed incremental rental in the model. The tank option still loses under the tested downside cases. Validate rental inclusions, customer volumes and project costs.

ILLUSTRATIVE high-demand/low-domestic-delivery case. NPVs are incremental versus existing assets at 10%, pre-tax. Each shock applied separately: capex +25%; no chargeable excess services; Vopak gateway capture 80%→60%; captured Lesedi flow −10%. No probabilities. Full numeric outputs and cash flows retained.

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Estimated working tankage supports a handling sensitivity of 3.50 and 2.98bn litres at two turns

Site	Working estimate	Low turns	Base turns	High turns
Durban	146,000 m³	1.75bn L 1 turns/month	3.50bn L 2 turns/month	5.26bn L 3 turns/month
Lesedi	124,000 m³	1.49bn L 1 turns/month	2.98bn L 2 turns/month	3.57bn L 2.4 turns/month

Durban: estimated petrol/diesel subset, not all gross tankage. Lesedi: 88,000 m³ documented plus 36,000 estimated. Handling sensitivities are not measured throughput or guaranteed operating limits.

Registered terminal_site_assumptions.csv. Capture cases hold base turns at two/month; the investment illustration separately tests 1.25 / 1.50 turns with the same Lesedi starting working capacity. Product allocation, access, peaks and downtime require validation.

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